

Two treatment comparisons

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice conducting paired t -tests, interpreting confidence intervals, and thinking carefully about how data structure affects analysis.

In this assignment, you will analyze data from a strip-trial evaluating whether a cereal rye cover crop reduces corn yield. Corn was planted following either:

- No cover crop (Control)
- Cereal rye terminated prior to planting.

Strips were paired within blocks across the field to control for spatial variability. Your task is to determine whether the rye treatment affected corn yield.

You can see an example of the trial design.

Block	12	Control	Cereal Rye
	11	Cereal Rye	Control
	10	Control	Cereal Rye
	9	Cereal Rye	Control
	8	Control	Cereal Rye
	7	Cereal Rye	Control
	6	Cereal Rye	Control
	5	Control	Cereal Rye
	4	Control	Cereal Rye
	3	Control	Cereal Rye
	2	Cereal Rye	Control
	1	Control	Cereal Rye

Treatment

Cereal Rye

Control

Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

Question 1

1 point

Load the dataset “data/cover-crop.csv” and assign it to an object called rye.

Inspect the first six rows using head().

```
library(ggplot2)
rye <- read.csv('./data/cover-crop.csv')
head(rye)
```

```
##   Block Treatment Yield
## 1     1   Control 203.0
## 2     1 Cereal Rye 192.4
## 3     2   Control 180.5
## 4     2 Cereal Rye 191.0
## 5     3   Control 188.1
## 6     3 Cereal Rye 178.2
```

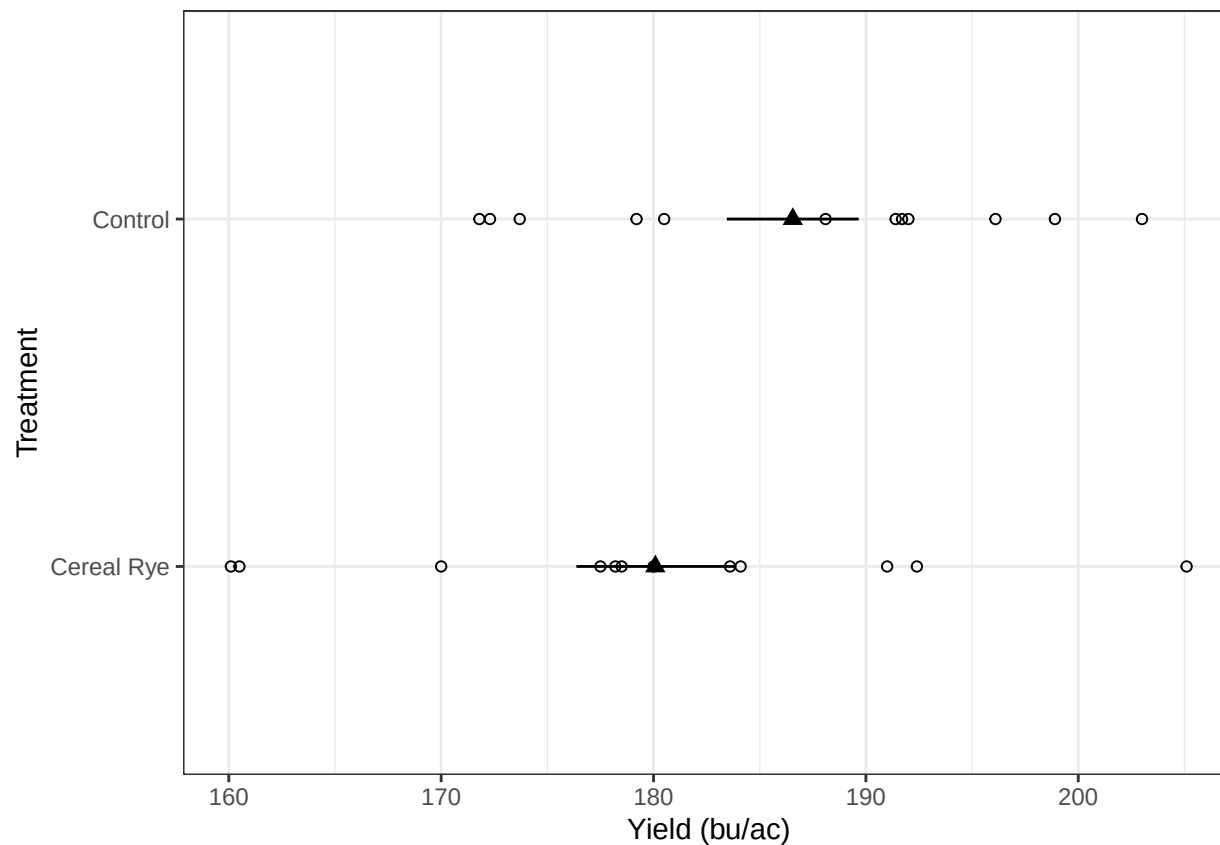
Question 2

2 points

Before analyzing the data, let's visualize it. Build a plot in which you show the yield observations of both treatments. This next part is optional but, how about adding the mean for each treatment and the standard deviation?

hint: there's a graph like this in the lecture HTML. Also, pay attention to the column names. Capitalization matters in R.

```
ggplot(rye)+
  geom_point(aes(x = Yield, y = Treatment), pch = 1)+
  labs(x = 'Yield (bu/ac)', y = 'Treatment')+
  stat_summary(aes(x = Yield, y = Treatment),
               pch = 17,
               fun.data = 'mean_se')+
  theme_bw()
```



Question 3

3 points

Because this is a paired design, we will analyze the difference within each block.

1. Reshape the dataset into wide format so that each treatment has its own column.
2. Compute a new column called diff representing: Control yield-Rye yield

```
rye_wide <- reshape(rye,
  idvar = 'Block',
  timevar = 'Treatment',
  direction = 'wide')
rye_wide
```

```
##      Block Yield.Control Yield.Cereal Rye
## 1         1         203.0         192.4
## 3         2         180.5         191.0
## 5         3         188.1         178.2
## 7         4         171.8         160.1
## 9         5         192.0         178.5
## 11        6         173.7         160.5
## 13        7         172.3         180.0
```

```
## 15      8      191.4      177.5
## 17      9      198.9      184.1
## 19     10      179.2      170.0
## 21     11      196.1      205.1
## 23     12      191.7      183.6
```

```
rye_wide$diff <- rye_wide$Yield.Control - rye_wide$Yield.Cereal Rye
rye_wide
```

```
##      Block Yield.Control Yield.Cereal Rye  diff
## 1         1      203.0      192.4  10.6
## 3         2      180.5      191.0 -10.5
## 5         3      188.1      178.2   9.9
## 7         4      171.8      160.1  11.7
## 9         5      192.0      178.5  13.5
## 11        6      173.7      160.5  13.2
## 13        7      172.3      180.0  -7.7
## 15        8      191.4      177.5  13.9
## 17        9      198.9      184.1  14.8
## 19       10      179.2      170.0   9.2
## 21       11      196.1      205.1  -9.0
## 23       12      191.7      183.6   8.1
```

Question 4

3 points

Using the vector of differences: 1. Compute the mean difference 2. Compute the standard error 3. Compute the t-value that we can use to construct a 95% confidence interval

```
xbar <- mean(rye_wide$diff)
s <- sd(rye_wide$diff) / sqrt(length(rye_wide$diff))
t.val <- qt(c(0.025, 0.975), nrow(rye_wide) - 1)

xbar
```

```
## [1] 6.475
```

```
s
```

```
## [1] 2.771175
```

```
t.val
```

```
## [1] -2.200985  2.200985
```

Question 5

1 point

Construct a 95% confidence interval

```
ci <- xbar + (s * t.val)
ci
```

```
## [1] 0.3756842 12.5743158
```

Question 6

2 points

Interpret your confidence interval. Does the interval include zero? What does this tell us about whether cereal rye reduces corn yield?

Question 7

2 points

Use the built-in `t.test()` function in R to conduct a paired *t*-test comparing rye and control treatments.

Report: 1. The *t*-statistic 2. The p-value 3. The 95% confidence interval

```
t.test(rye_wide$Yield.Control,
       rye_wide$`Yield.Cereal Rye`,
       paired = TRUE)
```

```
##
## Paired t-test
##
## data: rye_wide$Yield.Control and rye_wide$`Yield.Cereal Rye`
## t = 2.3366, df = 11, p-value = 0.03941
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 0.3756842 12.5743158
## sample estimates:
## mean difference
## 6.475
```

Question 8

1 point

Write a brief interpretation of the *t*-test output. Things worth mentioning are the hypothesis you tested, the p-value, and whether you reject the null hypothesis.